

Q1 JEE Main 2020 - 7 Jan (Morning)

If distance between the foci of an ellipse is 6 and distance between its directrices is 12, then length of its latus rectum is

- (A) 4
- (B) $3\sqrt{2}$
- (C) 9
- (D) $2\sqrt{2}$

Q2 JEE Main 2020 - 7 Jan (Evening)

If $3x + 4y = 12\sqrt{2}$ is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$ for some $a \in R$, then the distance between the foci of the ellipse is:

- (A) $2\sqrt{5}$
- (B) $2\sqrt{3}$
- (C) $2\sqrt{7}$
- (D) 4

Q3 JEE Main 2020 - 8 Jan (Morning)

Let the line $y = mx$ and the ellipse $2x^2 + y^2 = 1$ intersect at a point P in the first quadrant. If the normal to this ellipse at P meets the co-ordinate axes at $\left(-\frac{1}{3\sqrt{2}}, 0\right)$ and $(0, \beta)$, then β is equal to

- (A) $\frac{2}{\sqrt{3}}$
- (B) $\frac{2\sqrt{2}}{3}$
- (C) $\frac{\sqrt{2}}{3}$
- (D) $\frac{2}{3}$

Q4 JEE Main 2020 - 9 Jan (Morning)

If e_1 and e_2 are eccentricities of the ellipse $\frac{x^2}{18} + \frac{y^2}{4} = 1$ and the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ respectively and if the point (e_1, e_2) lies on ellipse $15x^2 + 3y^2 = k$. Then find value of k

- (A) 14
- (B) 15

(C) 16

(D) 17

Q5 JEE Main 2020 - 9 Jan (Evening)

The length of the minor axis (along y -axis) of an ellipse in the standard form is $\frac{4}{\sqrt{3}}$. If this ellipse touches the line, $x + 6y = 8$; then its eccentricity is :

(A) $\frac{1}{3}\sqrt{\frac{11}{3}}$

(B) $\sqrt{\frac{5}{6}}$

(C) $\frac{1}{2}\sqrt{\frac{11}{3}}$

(D) $\frac{1}{2}\sqrt{\frac{5}{3}}$

Q6 JEE Main 2020 - 3 Sep (Evening)

Let e_1 and e_2 be the eccentricities of the ellipse, $\frac{x^2}{25} + \frac{y^2}{b^2} = 1 (b < 5)$ and the hyperbola, $\frac{x^2}{16} - \frac{y^2}{b^2} = 1$ respectively satisfying $e_1 e_2 = 1$. If α and β are the distances between the foci of the ellipse and the foci of the hyperbola respectively, then the ordered pair (α, β) is equal to :

(A) (8,10)

(B) $\left(\frac{24}{5}, 10\right)$

(C) $\left(\frac{20}{3}, 12\right)$

(D) (8,12)

Q7 JEE Main 2020 - 4 Sep (Morning)

Let $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b)$ be a given ellipse, length of whose latus rectum is 10. If its eccentricity is the maximum value of the function, $\phi(t) = \frac{5}{12} + t - t^2$, then $a^2 + b^2$ is equal to :

(A) 135

(B) 116

(C) 126

(D) 145

Q8 JEE Main 2020 - 4 Sep (Evening)

Let $x = 4$ be a directrix to an ellipse whose centre is at the origin and its eccentricity is $\frac{1}{2}$. If $P(1, \beta)$, $\beta > 0$ is a point on this ellipse, then the equation of the normal to it at P is

- (A) $7x - 4y = 1$
- (B) $4x - 2y = 1$
- (C) $4x - 3y = 2$
- (D) $8x - 2y = 5$

Q9 JEE Main 2020 - 5 Sep (Morning)

If the co-ordinates of two points A and B are $(\sqrt{7}, 0)$ and $(-\sqrt{7}, 0)$ respectively and P is any point on the conic, $9x^2 + 16y^2 = 144$, then $PA + PB$ is equal to :

- (A) 9
- (B) 16
- (C) 6
- (D) 8

Q10 JEE Main 2020 - 5 Sep (Morning)

If the point P on the curve, $4x^2 + 5y^2 = 20$ is farthest from the point $Q(0, -4)$, then PQ^2 is equal to :

- (A) 29
- (B) 48
- (C) 21
- (D) 36

Q11 JEE Main 2020 - 6 Sep (Morning)

Which of the following points lies on the locus of the foot of perpendicular drawn upon any tangent to the ellipse, $\frac{x^2}{4} + \frac{y^2}{2} = 1$ from any of its foci?

- (A) (1, 2)
- (B) $(-2, \sqrt{3})$
- (C) $(-1, \sqrt{3})$
- (D) $(-1, \sqrt{2})$

Q12 JEE Main 2020 - 6 Sep (Evening)

If the normal at an end of a latus rectum of an ellipse passes through an extremity of the minor axis, then the eccentricity e of the ellipse satisfies :

(A) $e^2 + 2e - 1 = 0$

(B) $e^2 + e - 1 = 0$

(C) $e^4 + 2e^2 - 1 = 0$

(D) $e^4 + e^2 - 1 = 0$

Answer Key

Q1 (B)

Q2 (C)

Q3 (C)

Q4 (C)

Q5 (C)

Q6 (A)

Q7 (C)

Q8 (B)

Q9 (D)

Q10 (D)

Q11 (C)

Q12 (D)

Q1

$$2ae = 6 \text{ and } \frac{2a}{e} = 12$$

$$\Rightarrow ae = 3 \text{ and } \frac{a}{e} = 6$$

$$\Rightarrow a^2 = 18$$

$$\Rightarrow b^2 = a^2 - a^2 e^2 = 18 - 9 = 9$$

$$\Rightarrow \text{L.R.} = \frac{2b^2}{a} = \frac{2 \times 9}{3\sqrt{2}} = 3\sqrt{2}$$

Q2

$$3x + 4y = 12\sqrt{2}$$

$$\Rightarrow 4y = -3x + 12\sqrt{2}$$

$$\Rightarrow y = -\frac{3}{4}x + 3\sqrt{2}$$

$$\text{condition of tangency } c^2 = a^2 m^2 + b^2$$

$$18 = a^2 \cdot \frac{9}{16} + 9$$

$$a^2 \cdot \frac{9}{16} = 9$$

$$a^2 = 16$$

$$1a = 4$$

$$e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4}$$

$$\therefore ae = \frac{\sqrt{7}}{4} \cdot 4 = \sqrt{7}$$

$$\therefore \text{ focus are } (\pm\sqrt{7}, 0)$$

$$\therefore \text{ distance between foci } = 2\sqrt{7}$$

Q3

Let P be (x_1, y_1)

Equation of normal at P is $\frac{x}{2x_1} - \frac{y}{y_1} = -\frac{1}{2}$

It passes through $\left(\frac{1}{3\sqrt{2}}, 0\right) \Rightarrow \frac{-1}{6\sqrt{2}x_1} = -\frac{1}{2}$

$$\Rightarrow x_1 = \frac{1}{3\sqrt{2}}$$

So $y_1 = \frac{2\sqrt{2}}{3}$ (as P lies in Ist quadrant)

$$\text{So } \beta = \frac{y_1}{x_1} = \frac{\sqrt{2}}{3}$$

Q4

$$e_1 = \sqrt{1 - \frac{4}{18}} = \sqrt{\frac{7}{9}} = \frac{\sqrt{7}}{3}$$

$$e_2 = \sqrt{1 + \frac{4}{9}} = \sqrt{\frac{13}{9}} = \frac{\sqrt{13}}{3}$$

$$15e_1^2 + 3e_2^2 = k \Rightarrow k = 15\left(\frac{7}{9}\right) + 3\left(\frac{13}{9}\right)$$

$$\therefore k = 16$$

Q5

$$2b = \frac{4}{\sqrt{3}} \Rightarrow b = \frac{2}{\sqrt{3}}$$

$$\text{Equation of tangent} = y = mx \pm \sqrt{a^2m^2 + b^2}$$

$$\text{comparing with } y = \frac{-x}{6} + \frac{4}{3}$$

$$m = \frac{-1}{6} \text{ and } a^2m^2 + b^2 = \frac{16}{9}$$

$$\Rightarrow \frac{a^2}{36} + \frac{4}{9} = \frac{16}{9}$$

$$\Rightarrow \frac{a^2}{36} = \frac{16}{9} - \frac{4}{9} = \frac{4}{9}$$

$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

$$e = \sqrt{1 - \frac{4}{3 \times 16}} = \sqrt{\frac{11}{12}}$$

Q6

$$\text{Equation of ellipse: } \frac{x^2}{25} + \frac{y^2}{b^2} = 1$$

$$e_1 = \sqrt{1 - \frac{b^2}{25}}$$

$$\text{Equation of hyperbola: } \frac{x^2}{16} - \frac{y^2}{b^2} = 1 \text{ then } e_2 = \sqrt{1 + \frac{b^2}{16}}$$

$$\therefore e_1 \cdot e_2 = 1$$

$$\left(\frac{25-b^2}{25} \right) \left(\frac{16+b^2}{16} \right) = 1$$

$$\Rightarrow b^2 (b^2 - 9) = 0$$

$$\Rightarrow b = 3$$

$$e_1 = \frac{4}{5} \text{ and } e_2 = \frac{5}{4}$$

$$\therefore \text{Distance between focii of ellipse} = \alpha = 8$$

$$\text{Distance between focii of hyperbola} = \beta = 10$$

Q7

$$\therefore \text{Ellipse } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b)$$

$$\text{Given } \frac{2b^2}{a} = 10 \Rightarrow b^2 = 5a$$

$$\text{Now } \phi(t) = \frac{5}{12} + t - t^2$$

$$\phi'(t) = 1 - 2t = 0 \Rightarrow t = \frac{1}{2}$$

$$\phi''(t) = -2 < 0 \Rightarrow \text{maximum}$$

$$\Rightarrow \phi(t)_{\max} = \frac{5}{12} + \frac{1}{2} - \frac{1}{4} = \frac{8}{12} = \frac{2}{3} = e \text{ (given)}$$

$$\text{Now } b^2 = a^2 (1 - e^2)$$

$$5a = a^2 \left(1 - \frac{4}{9}\right)$$

$$5a = \frac{5a^2}{9} \Rightarrow a^2 - 9a = 0$$

$$\Rightarrow a = 9 \Rightarrow a^2 = 81 \Rightarrow b^2 = 45$$

$$\Rightarrow a^2 + b^2 = 81 + 45 = 126$$

Q8

$$\text{Given } x = 4$$

$$\Rightarrow \pm \frac{a}{e} = 4$$

$$\Rightarrow a = 4e$$

$$\Rightarrow a = 4 \times \frac{1}{2} = 2$$

$$\Rightarrow a = 2$$

$$\text{Now, } b^2 = a^2 (1 - e^2)$$

$$\Rightarrow b^2 = 4 \left(1 - \frac{1}{4}\right) = 4 \times \frac{3}{4} = 3$$

$$\text{So, equation } \frac{x^2}{4} + \frac{y^2}{3} = 1$$

$$\Rightarrow 3x^2 + 4y^2 = 12$$

$$\text{Now } P(1, \beta) \text{ lies on it}$$

$$\Rightarrow 3 + 4\beta^2 = 12$$

$$\Rightarrow 4\beta^2 = 9$$

$$\Rightarrow \beta = \frac{3}{2}$$

$$\text{So, } P\left(1, \frac{3}{2}\right)$$

$$\text{Now differentiating (i)}$$

$$6x + 8y \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-3x}{4y}$$

Ellipse

JEE Main 2020 Chapterwise

Hints & Solutions

Mathematics

$$\Rightarrow \frac{dy}{dx} \Big| \left(1, \frac{3}{2}\right) = -\frac{1}{2}$$

So, slope of normal = 2

So, equation of normal: $y - \frac{3}{2} = 2(x - 1)$

$$\Rightarrow 2y - 3 = 4x - 4$$

$$\Rightarrow 4x - 2y = 1$$

Q9

$$E : \frac{x^2}{16} + \frac{y^2}{9} = 1$$

$(\pm\sqrt{7}, 0)$ are the foci of given ellipse. So for any point P on it; $PA + PB = 2a$

$$\Rightarrow PA + PB = 2(4) = 8$$

Q10

$$\text{Curve is } C \equiv \frac{x^2}{5} + \frac{y^2}{4} = 1$$

Let a point on curve be $(\sqrt{5} \cos \theta, 2 \sin \theta)$

$$PQ^2 = (\sqrt{5} \cos \theta)^2 + (-4 - 2 \sin \theta)^2$$
$$= 5 \cos^2 \theta + 4 \sin^2 \theta + 16 + 16 \sin \theta$$

$$\Rightarrow PQ^2 = 21 + 16 \sin \theta - \sin^2 \theta$$

$$= 21 + 64 - (\sin \theta - 8)^2$$

$$= 85 - (\sin \theta - 8)^2$$

For PQ^2 to be maximum $\sin \theta = 1$

$$PQ_{\max}^2 = 85 - 49 = 36$$

Q11

Locus will be $x^2 + y^2 = 4$ (auxiliary circle) $(-1, \sqrt{3})$ satisfies the given equation (Property of ellipse is that locus of foot of perpendicular from any foci to tangent lies on auxiliary circle of ellipse)

Q12

Normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $\left(ae_1, \frac{b^2}{a}\right)$ is

$$x - ey = \frac{e(a^2 - b^2)}{a}$$

$\therefore (0, -b)$ lies on it, so

Ellipse

Hints & Solutions

JEE Main 2020 Chapterwise

Mathematics

$$be = \frac{e(a^2 - b^2)}{a}$$

$$\Rightarrow ab = a^2 e^2$$

$$\Rightarrow b = ae^2 \Rightarrow \frac{b^2}{a^2} = e^4$$

$$1 - e^2 = e^4 \Rightarrow e^4 + e^2 - 1 = 0$$